Saurav Koduri

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Education

University of Illinois Urbana-Champaign
Major: BS, Electrical Engineering; Cumulative GPA: 3.91/4.00 — *Class of 2028*

Experience

Sage Program Researcher — Summer 2026
Argonne National Lab
- Built an edge-AI wildlife detection pipeline using YOLO11, BioCLIP 2, and Gemma 3 for detection, species identification, and risk scoring.
- Developed multi-frame animal tracking and automated alerts while preventing duplicate messages.
- Containerized the application for NSF Sage edge-computing infrastructure using Docker, PyWaggle, and Sage job manifests, integrating camera-image ingestion, structured detection publishing, persistent tracker state, and fault-tolerant CSV and annotated-image outputs.

Robotics Engineer — Fall 2025 – Present
UIUC SigRobotics
- Designing a custom four-layer controller and power-distribution PCB in KiCad for a five-finger robotic hand using an ESP32-S3 and 10 DYNAMIXEL XL330-M288-T actuators.
- Implementing a shared 3.3V TTL half-duplex communication interface, native USB-C programming, boot/reset circuitry, debugging access, ESD protection, and expansion interfaces.
- Engineering a protected 3S battery power architecture with separate logic and motor rails, including a high-current 5V converter targeting 15–16A continuous output, power budgeting, component selection, and thermal/copper planning.

Machine Learning Intern — Summer 2025
Sizzle.gg
- Wrote Python scripts to build a dataset of over 100,000 gameplay images across multiple games
- Building and refining an image classification model to automatically detect the game being played on screen. The model is presently detecting 7 games and expanding it to detect over 50 games.
- Migrated video game streaming data from MongoDB database to PostgreSQL database by building a new schema for PostgreSQL and configuring data structures to preserve integrity.

Project Associate — Spring 2026 – Present
Illinois Enactus
- Building a platform to help reduce language barriers for the Q'anjob'al community in Champaign
- Developed platform features focused on multilingual accessibility by applying user-centered design, language support workflows, and technical problem-solving to improve communication

Projects

Autonomous Agricultural Tractor System — Summer 2026
- Building AI models on ROS based tractor vehicle to autonomously drive through pecan farm, upgrading systems from YOLO to Alpamayo models to allow for increased accuracy
- Migrating all packages from ROS 1 to ROS 2 in order to improve framework for AI models
- Creating Isaac sim simulation of tractor system to improve testing system and enhance sensor system

Resistive-Key Vehicle Safety System — Spring 2026
- Created an analog authentication circuit that converts a resistor-based key value into a target voltage
- Integrated comparator circuits and AND-gate logic to validate the key voltage, then implemented a three-second delay circuit to slow repeated resistance attempts to reduce successful hacker attempts

Technical Skills

Software/Hardware: C/C++, Python, Linux, SQL, SolidWorks, Matlab, ROS 2, Isaac Sim, KiCad, PyTorch
Courses: Digital Systems, Analog Signal Processing, Computer Systems and Programming, Data Structures and Algorithms